Data Flow Diagram

Date	06 July 2026
Team ID	TMID-SL-2026-01
Project Name	SmartLender - Applicant Credibility Prediction for Loan Approval
Maximum Marks	2 Marks

Create a Data Flow Diagram (DFD) to show how data moves through your system — between external entities, processes, and data stores.

Symbol	Name	Description
Oval	External Entity	Applicant, Credit Officer — sources/consumers of data outside the system
Numbered rectangle	Process	An activity that transforms data (e.g., Validate Input, Predict Eligibility)
Open rectangle	Data Store	Where data is held (Dataset, rdf.pkl, scale1.pkl)
Labeled arrow	Data Flow	Movement of data between entities, processes and stores

Level-0 (Context) Diagram

Applicant / Credit Officer → (1.0) Smart Lender System → Approved / Rejected result → back to Applicant / Credit Officer

Level-1 Diagram

#	Process	Input From	Output To
1.0	Collect applicant details (predict.html form)	Applicant / Credit Officer	1.0 Validate Input
2.0	Validate & encode input (app.py)	1.0	3.0 Scale Features
3.0	Scale features (scale1.pkl)	2.0	4.0 Predict Eligibility
4.0	Predict eligibility (rdf.pkl — XGBoost model)	3.0	5.0 Render Result
5.0	Render result (submit.html)	4.0	Applicant / Credit Officer

Data stores referenced: **Dataset/loan_prediction.csv** (training data), **Flask/rdf.pkl** (trained model), **Flask/scale1.pkl** (fitted scaler).